power

File: power.kicad_sch

sensors

File: sensors.kicad_sch

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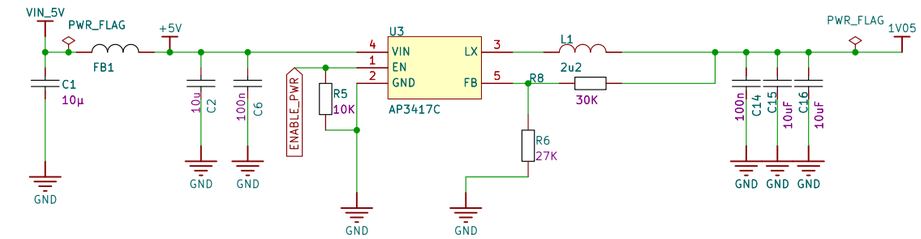
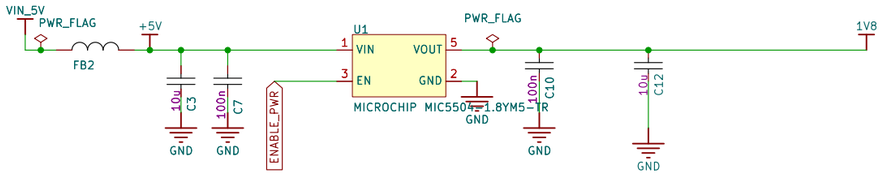
File: sensor_board.kicad_sch

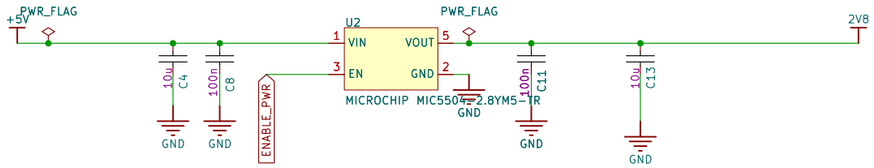
Title:

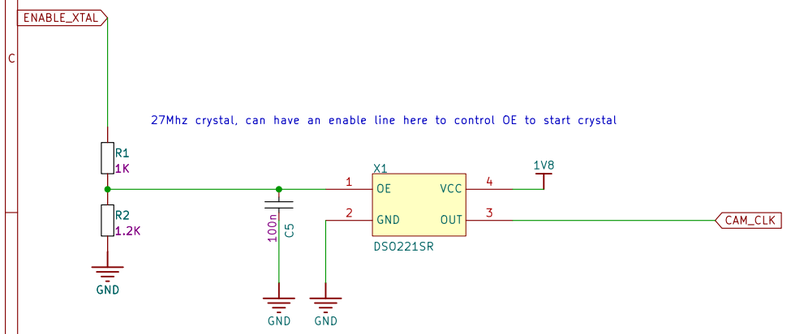
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 1/3



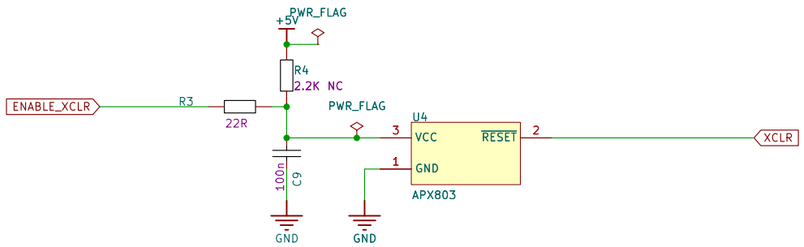


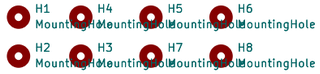
2.8V (VANA) Analog Pixel Array Low (~50mA) LDO Noise Sensitive. Any ripple here creates visible lines in the image. Efficiency doesn't matter because current is low.
1.8V (VIF) I/O Interface Low (~30mA) LDO Interface Level. Keeps signals clean for I2C communication.
1.05V (VDIG) Digital Logic Core High (~250mA+) Buck Power Hungry. Dropping 3.3V or 5V to 1.05V via LDO generates too much heat. A Buck runs cool.





27Mhz crystal, can have an enable line here to control OE to start crystal





1. The Circuit Logic (Weak vs. Strong)
The APX803 is a voltage supervisor that monitors the voltage on its own VCC pin. If VCC drops below a specific threshold (e.g., 2.93V or 4.38V), it asserts the RESET signal (pulls it LOW).
5V via 2.2kΩ (2200Ω): This acts as a "Weak" source. It tries to pull the VCC pin up to 5V, but it has high resistance, so it cannot supply much current or fight against a stronger signal.
ENABLE via 22Ω: This acts as a "Strong" driver. Because 22Ω is 100 times smaller than 2200Ω, the voltage on the ENABLE line will dominate and "overpower" the 5V source.

2. How Control Works
The voltage at the APX803's VCC pin will essentially follow the ENABLE signal voltage.
When ENABLE is LOW (0V / Ground):
The 22Ω resistor pulls the VCC pin down to approximately 0.05V (acting as a voltage divider with the 2.2kΩ resistor).
Result: The APX803 sees this as a "Power Failure" (Voltage < Threshold). It immediately Asserts RESET (drives the Reset pin LOW), holding the target system in a reset state.

When ENABLE is HIGH (e.g., 3.3V or 5V):
The ENABLE signal drives the VCC pin up to a valid logic level (e.g., ~3.3V or ~5V). The 2.2kΩ resistor just limits the current flow between the 5V rail and the ENABLE rail.
Result: The APX803 sees valid power.[1][2] It waits for its internal delay timer (typically ~200ms) and then Releases RESET (lets the Reset pin go HIGH/Floating), allowing the system to run.

Sheet: /power/		
File: power.kicad_sch		
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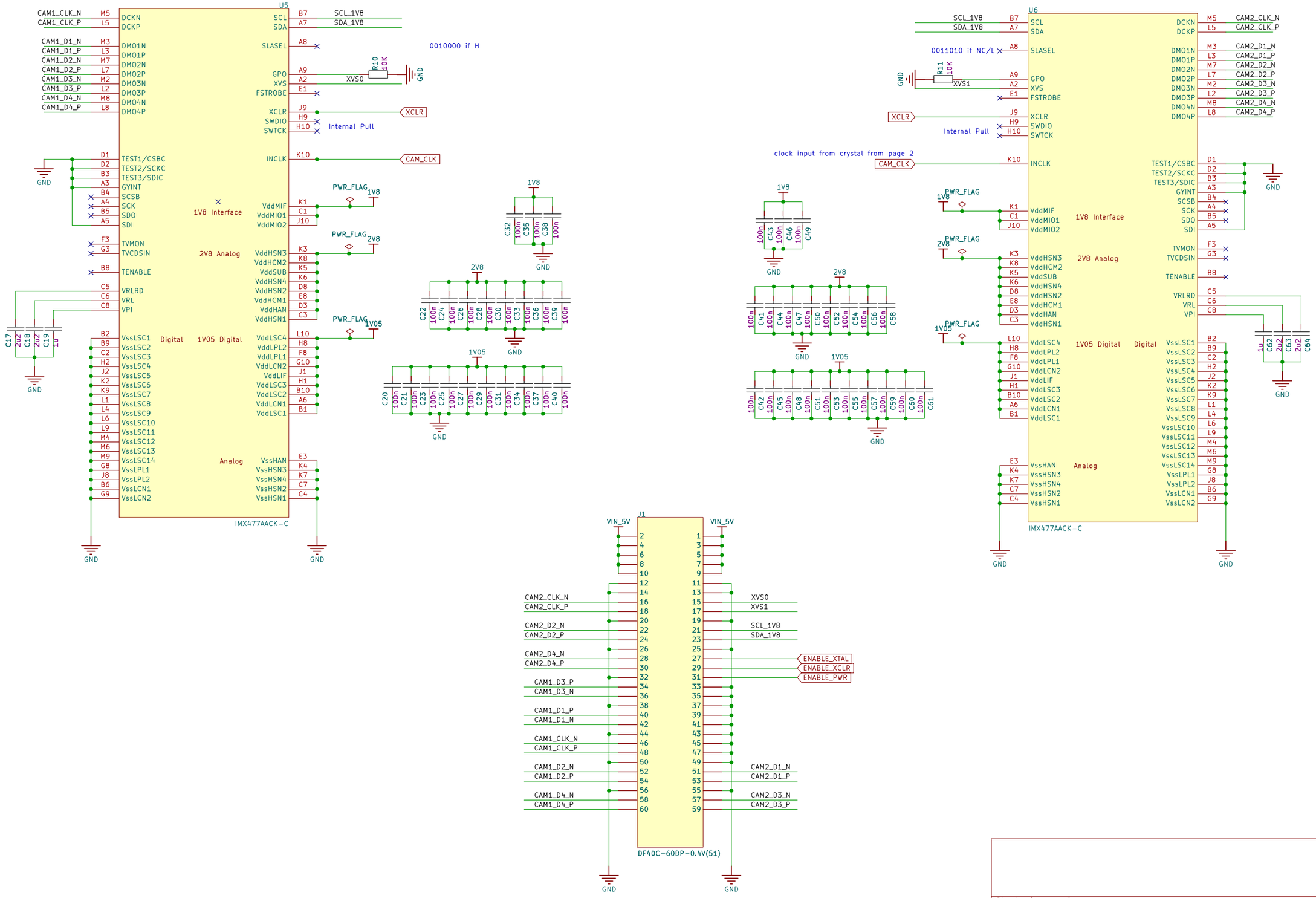
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 2/3



Sheet: /sensors/
File: sensors.kicad_sch

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Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 3/3